Application of Dynamic Time Warping and Soft-DTW in Time-Series Clustering
动态时间规整和软动态时间规整在时间序列聚类中的应用

a thesis

submitted to School of Mathematics & Physics

of Xi'an Jiaotong-Liverpool University

in partial fulfilment for the award of the degree of

Bsc Financial Mathematics

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April 29, 2025

Abstract

Time-series data plays a crucial role in fields such as finance, healthcare, and engineering, yet challenges such as variability, temporal misalignment, and nonlinearity complicate effective clustering. Dynamic Time Warping (DTW) provides flexible sequence alignment but suffers from non-differentiability, limiting its optimization applications. Soft-DTW addresses this limitation by introducing a smoothing parameter while preserving temporal robustness. In this study, the soft-DTW-based K-means cluster was applied to the weekly closing prices of Chinese A-share stocks across four sectors: Semiconductors, Banking, Real Estate Development, and Food Beverage, during 2023-2025, and the clustering performance was validated using the Silhouette Score, Within-Cluster Sum of Squares (WSS) and Davies-Bouldin Index (DBI). The results reveal strong intra-sector consistency and cross-sector dynamic similarities, demonstrating that Soft-DTW enhances the robustness and interpretability of financial time-series clustering.

时间序列数据在金融、医疗和工程等领域中具有重要作用，但其变异性、时间轴错位和非线性特性使得有效聚类面临挑战。动态时间规整（DTW）能够灵活对齐序列，但由于不可微分性限制了其在优化任务中的应用。Soft-DTW 通过引入平滑参数，实现了可微性并保留了时间弹性。本文采用基于 Soft-DTW 的 K-means 方法，对 2023 至 2025 年间中国 A 股半导体、银行、房地产开发和食品饮料四大行业的周收盘价序列进行聚类，并通过轮廓系数（Silhouette Score）、类内平方和（WSS）及戴维森堡丁指数（DBI）等指标验证了聚类效果。结果展示了行业内部的一致性与行业之间的动态相似性，验证了 Soft-DTW 在提升金融时间序列聚类稳健性与可解释性方面的有效性。

KEY WORDS: Time Series Clustering, Dynamic Time Warping, Stock clustering

关键词: 时间序列聚类，动态时间规整，股票聚类

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Chapter 1

Introduction

Recent developments in time-series clustering have achieved notable success in various areas, including network and system analysis, weather studies, social media, behavior analysis, trajectory data, biological research, and financial applications. Identifying meaningful patterns from large datasets calls for interdisciplinary efforts across fields like statistics, machine learning, data visualization, pattern recognition, and advanced computing[2].

Most mining methods leverage the concept of similarity between time series pairs. When working with time-series data, it is crucial to select an appropriate similarity measure to account for outliers, variations in amplitude, and distortions along the time axis. The efficiency and effectiveness of representation methods and similarity measures are primary objectives in handling time-series data [18]. Tornai et al. [17] emphasized that the distance between sequences, as a measure, plays a critical role in the quality of clustering algorithms. The choice of similarity measure can significantly impact the accuracy of such algorithms. Yahyaoui et al. [19] and Wang et al. [18] conducted comprehensive reviews of time-series measurement methods, categorizing them into four main types: synchronous measures (e.g., Euclidean distance and Manhattan distance), elastic measures (e.g., longest common subsequence and dynamic time warping (DTW)), pattern-based measures (e.g., spatial assembling distance), and threshold-based measures (e.g., threshold query-based similarity search).

In this report, we primarily focus on Dynamic Time Warping (DTW). DTW, as a geometry-driven method, is widely used for optimal alignment and similarity analysis of time series. It is particularly favored for its ability to handle variable-length sequences and its robustness to shifts and scaling along the time dimension[5, 8].

Dynamic Time Warping (DTW) has become an indispensable tool in the field of time series analysis. It is widely used across various domains, including multimedia retrieval, speech recognition, and financial forecasting, where it allows for precise alignment and similarity measurement of variable-length sequences[12]. For instance, in music information retrieval, DTW can synchronize audio tracks with corresponding musical scores, enabling advanced applications such as

real-time performance analysis[8]. The impact of DTW extends far beyond traditional applications, as it has significantly contributed to the development of multimodal data integration and machine learning frameworks. These innovations are crucial for tasks like cross-modal data retrieval, such as aligning video with text or analyzing the relationship between audio and motion sequences.[7]

In summary, DTW is not just a fundamental alignment algorithm; it has the potential to push the frontier of time series analysis in an increasingly data-driven world. It facilitates the seamless integration of data across different domains, such as music and motion. With variants like Soft-DTW Divergence overcoming traditional limitations, DTW is opening up new possibilities in real-time applications, personalized recommendations, and large-scale predictive modeling. By continuously optimizing and refining DTW-based methods, researchers and practitioners can leverage its unique advantages to solve complex time series problems, driving innovation and creating transformative impacts across industries.

Chapter 2

Literature Review

2.1 Overview of time series clustering methods

Time series clustering methods can be broadly categorized into three types: distance-based methods, model-based methods, and deep learning methods[1]. Distance-based methods, such as Euclidean distance and Dynamic Time Warping (DTW), classify sequences by measuring their similarity, with DTW being particularly effective for handling time-axis distortions [9, 4]. Model-based methods capture global patterns by fitting statistical models like ARMA or HMM, but their application is limited when dealing with large-scale datasets [10, 3]. Deep learning methods, such as CNN and LSTM, significantly improve classification performance by automatically extracting features and modeling temporal dependencies [9]. Overall, distance-based methods are efficient and simple, making them suitable for small datasets, while deep learning methods are better suited for more complex tasks.

2.2 Development and application of dynamic time warping

Dynamic Time Warping (DTW) is a widely used technique in time series analysis, initially developed for speech recognition and later expanded to fields such as music retrieval and motion analysis [16]. The fundamental idea of DTW is to align two time series through nonlinear warping, addressing time axis differences caused by variations in speed or rhythm [12]. To enhance its applicability and performance, researchers have introduced several variants, such as Local Weighted DTW (which adjusts the matching path cost by setting local weights) and Multiscale DTW (which improves efficiency by progressively matching from coarse to fine scales) [11].

DTW's application in time series classification has been extensively studied. For instance, in the music domain, DTW can be used for audio matching and synchronization, helping to identify the same musical segments across different performances [12]; in motion analysis, DTW supports similarity retrieval of motion capture data, even when there are significant differences in spatial and temporal structure [6]. Additionally, DTW is also applied in finance and medicine, where

it is used to compare time series patterns and identify specific anomalous events.

Despite its strong adaptability, DTW's high time complexity limits its use in large-scale datasets. To address this, researchers have developed several optimization methods, such as Constrained DTW (which restricts the search space by applying global constraints) and Subsequence DTW (which enables fast matching of local similar segments in long sequences) [12, 6]. These improvements not only enhance the algorithm's efficiency but also increase its robustness against data noise and distortions. Currently, research in DTW is mainly focused on two directions: one aims to improve DTW's speed[15], and the other explores new applications of DTW.

2.3 DTW and Soft DTW

However, DTW has some inherent limitations when used as a loss function: it is not differentiable everywhere within its domain, which can lead to local optima during optimization. Additionally, DTW is sensitive to noise, which may result in poor performance in certain applications [8, 14].

To overcome these limitations, Cuturi and Blondel introduced Soft-DTW in 2017 [8]. By replacing the traditional minimum operation in DTW with a soft minimum, Soft-DTW incorporates a probability distribution over all possible paths. This modification makes Soft-DTW differentiable everywhere, while maintaining the same computational complexity as traditional DTW, allowing it to be integrated into end-to-end differentiable optimization workflows. Soft-DTW has been successfully applied in various fields, such as audio-to-score alignment, video segmentation, and spatiotemporal sequence analysis [8].

Although Soft-DTW overcomes the non-differentiability issue of traditional DTW by introducing the soft minimum, making it suitable as a loss function in optimization problems, it still has some limitations. First, Soft-DTW is highly sensitive to the choice of the smoothing parameter , and different values of can significantly affect both the alignment results and the convergence of optimization [5]. Second, since Soft-DTW relies on dynamic programming, its computational complexity remains quadratic, which is inefficient when dealing with large-scale datasets [5].

To address these issues, researchers have proposed several improvements. For instance, Petitjean et al. [13] introduced a Soft-DTW-based method for calculating the mean time series (barycenter computation), which reduces the dependency on the smoothing parameter by optimizing the geometric structure. Additionally, improvements in optimization algorithms, such as using the more efficient L-BFGS method, can significantly enhance the computational efficiency of Soft-DTW on high-dimensional time series [5].

Another important direction for improvement is adjusting the path weight distribution in Soft-DTW to make it better suited for aligning non-uniform time series. These advancements not only expand the applicability of Soft-DTW but also enhance its performance in classification and clustering tasks.

Chapter 3

Methodology

3.1 Classic DTW

Dynamic Time Warping (DTW) is a commonly used algorithm for calculating the similarity between time series. Unlike traditional Euclidean distance, DTW can handle time series of different lengths by aligning their time axes for similarity calculation. The core idea of DTW is to align time series by non-linear transformations, allowing for the comparison of sequences with different time scales[12].

As shown in Figure 3.1[1], this image intuitively demonstrates the basic principle of Dynamic Time Warping (DTW). The cost matrix visualizes the Manhattan distance between two real-valued sequences X and Y, with dark colors representing regions of low cost and light colors indicating high-cost areas. The optimal alignment path traverses the valleys of low cost in the cost matrix, effectively minimizing the total cost and achieving the best alignment between the sequences.

Given two time series $X = (x_1, x_2, \ldots, x_m)$ and $Y = (y_1, y_2, \ldots, y_n)$, DTW finds the optimal alignment path A that minimizes the sum of the distances between corresponding points along this path. This alignment path is not a simple linear match, but allows stretching or compressing the time series to minimize the matching error.

An (N, M)-warping path (or simply referred to as warping path if N and M are clear from the context) is a sequence $p = (p_1, \ldots, p_L)$ with $p_\ell = (n_\ell, m_\ell) \in [1 : N] \times [1 : M]$ for $\ell \in [1 : L]$ satisfying the following three conditions.

(i) **Boundary condition**: $p_1 = (1, 1)$ and $p_L = (N, M)$.

(ii) **Monotonicity condition**: $n_1 \leq n_2 \leq \cdots \leq n_L$ and $m_1 \leq m_2 \leq \cdots \leq m_L$.

(iii) **Step size condition**: $p_{\ell+1} - p_\ell \in \{(1, 0), (0, 1), (1, 1)\}$ for $\ell \in [1 : L - 1]$.

[1]Figure3.1 from Müller et al., Information Retrieval for Music and Motion, 20007.

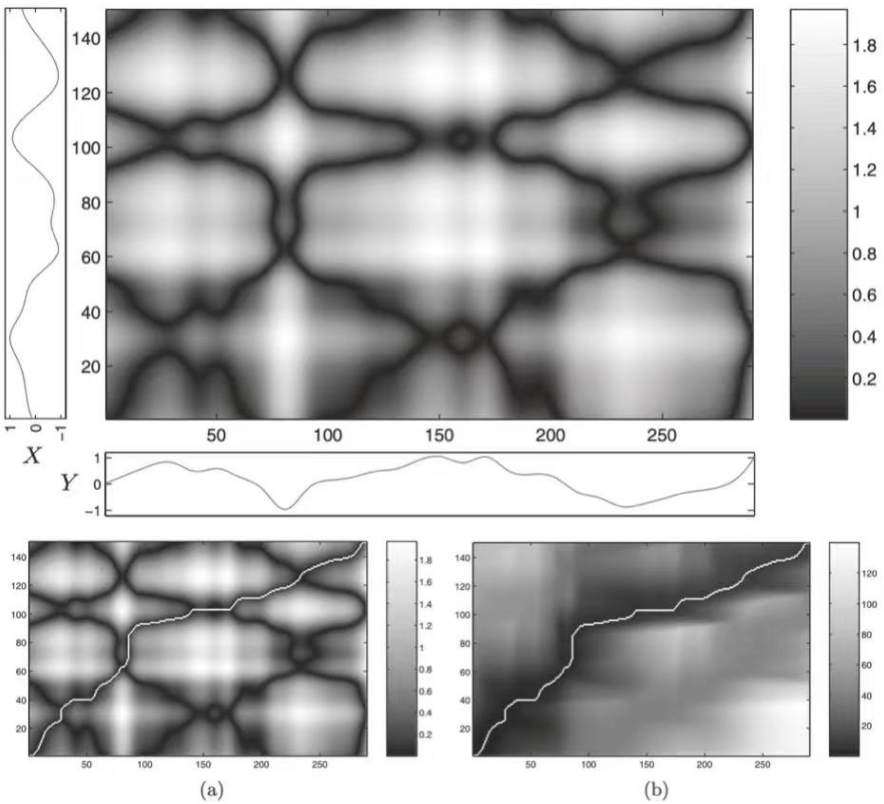

Figure 3.1: Cost matrix of the two real-valued sequences X (vertical axis) and Y (horizontal axis) using the Manhattan distance (absolute value of the difference) as local cost measure c. (a) Cost matrix C as in and (b) accumulated cost matrix D with optimal warping path p^* (white line).

The calculation process of DTW can be performed using dynamic programming, and the steps are as follows:

First, a cost matrix C is constructed to represent the distances between each pair of points in the time series. Euclidean distance is commonly used as the cost between points:

$$C_{ij} = \|x_i - y_j\|$$

where C_{ij} represents the distance between the i-th element in X and the j-th element in Y.

Dynamic programming is then used to calculate the optimal alignment path. A cumulative distance matrix D is built, where each element D_{ij} represents the minimum cumulative distance from $(1,1)$ to (i,j):

$$D_{ij} = C_{ij} + \min\left\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\right\}$$

where $D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}$ represent the minimum cumulative distances from the top, left, and top-left neighbors, respectively. This recursive process fills the matrix D, where D_{mn} represents the minimum distance along the optimal alignment path.

The optimal alignment path A is found by backtracking from $(1,1)$ to (m,n). Each element A_{ij} of the alignment path represents the matching of time point x_i with time point y_j.

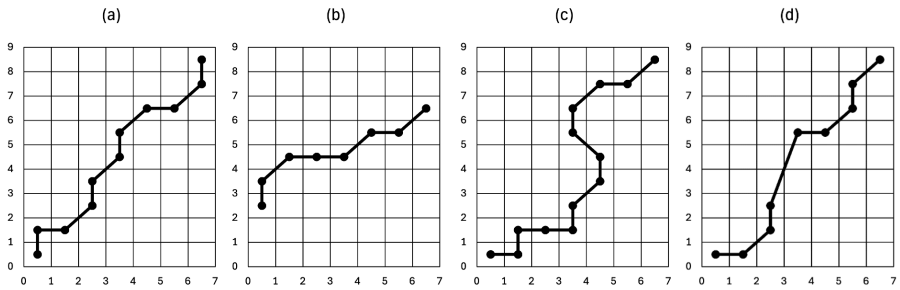

Figure 3.2: Illustration of paths of index pairs for some sequence X of length $N = 9$ and some sequence Y of length $M = 7$. (**a**) Admissible warping path satisfying the conditions (i), (ii), and (iii) of Definition 4.1. (**b**) Boundary condition (i) is violated. (**c**) Monotonicity condition (ii) is violated. (**d**) Step size condition (iii) is violated.

The Figure3.2 above illustrates of paths.

Finally, the result of DTW is the total cost D_{mn}, which represents the similarity between the time series X and Y. A smaller value means the sequences are more similar.

3.2 Soft-DTW

However, traditional Dynamic Time Warping (DTW) has the following limitations: non-differentiability and reliance on the uniqueness of the alignment path. Due to the discontinuity of the hard minimum operation, DTW cannot be directly used in gradient-based optimization methods. Moreover, DTW considers only a single optimal alignment path while ignoring other possible paths, making it highly sensitive to noise and outliers[8].

In Soft-DTW, we first define two matrices:

- The local cost matrix $C \in \mathbb{R}^{m \times n}$, where C_{ij} denotes the distance between the i-th element of sequence X and the j-th element of sequence Y;
- The alignment matrix $A \in \{0, 1\}^{m \times n}$, where each A_{ij} indicates whether (x_i, y_j) belongs to a given alignment path.

The core improvement of Soft-DTW lies in replacing the hard minimum ($\min$) in traditional DTW with a soft minimum ($\min_\gamma$). The soft minimum is defined as:

$$\min_\gamma \{a_1, \ldots, a_n\} = -\gamma \log \sum_{i=1}^{n} \exp\left(-\frac{a_i}{\gamma}\right),$$

where:

- $\gamma > 0$ is a smoothing parameter controlling the softness of the operation;
- as $\gamma \to 0$, the soft minimum converges to the traditional hard minimum;
- as $\gamma \to \infty$, the soft minimum approaches the arithmetic mean.

Using these definitions, the Soft-DTW loss between two sequences is formulated as:

$$\text{Soft-DTW}(C) = -\gamma \log \sum_{A \in A(m,n)} \exp\left(-\frac{\langle A, C\rangle}{\gamma}\right),$$

where $\langle A, C\rangle = \sum_{i,j} A_{ij} C_{ij}$ denotes the total cost along the alignment path A.

The Gibbs distribution over all possible alignment paths A is defined as:

$$\mathbb{P}_\gamma(A) = \frac{\exp\left(-\frac{\langle A,C\rangle}{\gamma}\right)}{Z}, \quad \text{with} \quad Z = \sum_{A' \in A(m,n)} \exp\left(-\frac{\langle A', C\rangle}{\gamma}\right).$$

This distribution assigns higher probabilities to lower-cost paths and lower probabilities to higher-cost paths, with the smoothing parameter γ controlling the sharpness of the distribution.

The recurrence relation for computing Soft-DTW via dynamic programming is given by:

$$r_{i,j} = \Delta_{i,j} + \min_\gamma \{r_{i-1,j-1}, r_{i-1,j}, r_{i,j-1}\},$$

where $\Delta_{i,j}$ is the element of the cost matrix C at position (i, j).

The initial conditions are:

$$r_{0,0} = 0, \quad r_{i,0} = r_{0,j} = +\infty \quad (\text{for } i > 0 \text{ or } j > 0).$$

The time complexity and space complexity of Soft-DTW are both $O(mn)$, similar to classical DTW. Due to smoothing operations, numerical stabilization techniques such as log-sum-exp are often necessary.

One major advantage of Soft-DTW is its differentiability. The gradient of Soft-DTW with respect to C can be computed as:

$$\nabla_C \text{Soft-DTW}(C) = \mathbb{E}_\gamma[A; C],$$

where $\mathbb{E}_\gamma[A; C]$ denotes the expected alignment matrix under the Gibbs distribution, providing a weighted average over all possible alignment paths.

The smoothing parameter γ offers flexible control over optimization:

- Smaller γ approximates the hard minimum but may cause numerical instability;
- Larger γ provides smoother optimization landscapes, helping to avoid local minima.

Furthermore, Soft-DTW degenerates into classical DTW as $\gamma \to 0$ and converges to the mean cost over all paths as $\gamma \to \infty$.

3.3 K-means

K-means clustering is a widely used unsupervised learning algorithm that partitions a dataset into k clusters by minimizing intra-cluster variance. Traditionally, it uses Euclidean distance to

measure similarity between data points and centroids. However, for time-series data, Euclidean distance often fails to capture sequence misalignments and temporal variations.

To address this, researchers have introduced the use of **Dynamic Time Warping (DTW)** and **Soft-DTW** as distance measures in K-means. Unlike DTW, which is non-differentiable, Soft-DTW provides a differentiable distance function, making it suitable for gradient-based optimization.

The classical K-means objective is:

$$\min_{\{\mu_j\}_{j=1}^k} \sum_{i=1}^n \min_{j=1,\dots,k} \|x_i - \mu_j\|^2$$

In Soft-DTW K-means, this becomes:

$$\min_{\{\mu_j\}_{j=1}^k} \sum_{i=1}^n \min_{j=1,\dots,k} \mathrm{SDTW}_\gamma(x_i, \mu_j)$$

where $\mathrm{SDTW}_\gamma(x_i, \mu_j)$ denotes the Soft-DTW distance between time series x_i (an input sequence) and μ_j (the corresponding cluster center sequence), and γ is the smoothing parameter controlling the softness of alignment.

Each cluster center μ_j, known as a *barycenter*, is updated not by averaging, but by solving the following optimization problem:

$$\mu_j^* = \arg\min_{\mu} \sum_{x_i \in C_j} \mathrm{SDTW}_\gamma(x_i, \mu)$$

This optimization is solved using methods such as L-BFGS. The gradient of the Soft-DTW loss with respect to the barycenter μ can be computed from the expected alignment matrix derived from the Gibbs distribution over all warping paths:

$$\nabla_\mu \mathrm{SDTW}_\gamma(x_i, \mu) = \mathbb{E}_\gamma[A; C]$$

By using Soft-DTW, the K-means algorithm becomes more robust to time warps, noise, and misaligned patterns. It is highly effective for clustering real-world time-series data such as physiological signals, financial prices, or movement trajectories.

Chapter 4

Data Illustration

This section aims to demonstrate the process and results of clustering closing prices of stocks in the dataset. A dataset was constructed by combining stock information from multiple sectors. After that, data cleaning and standardization were performed, followed by K-means clustering using Soft-DTW as the distance metric.

The stock data used in this study was provided by Baostock, while the sector classification was obtained from Eastmoney, accessed through the AKShare platform. The research focuses on weekly A-share data from the year 2023 to 2025. Each stock in the dataset contains 101 observations.

4.1 Data Description and Preprocessing

In this study, we use historical stock closing price data obtained from Baostock, an open-source financial data interface, and industry classification data from Eastmoney, accessed via the AKShare platform. The dataset is constructed based on four major sectors in the Chinese A-share market: Semiconductors, Banking, Real Estate Development, and Food & Beverage.

The selection of these four sectors is based on their distinct economic characteristics and market behaviors. The semiconductor sector represents technology-driven and high-volatility stocks; banking reflects stable, policy-sensitive financial entities; real estate development includes cyclical and asset-heavy firms; and the food and beverage industry typically features consumer defensive stocks with stable cash flows. By combining sectors with contrasting behaviors, this

Industry	**Mean**	**Std. Dev.**	**Max**	**Min**	**Median**	**Range**
Semiconductors	51.9527	54.1063	430.64	7.84	33.53	422.80
Real Estate Dev.	5.6048	3.9693	32.86	1.16	4.23	31.70
Banking	7.1330	6.0501	41.05	1.63	5.475	39.42
Food & Beverage	18.5193	20.0584	173.35	2.23	13.645	171.12

Table 4.1: Descriptive statistics of closing prices by industry

study aims to evaluate the clustering performance of Soft-DTW under heterogeneous time series patterns.

The time frequency chosen is weekly to smooth out daily volatility while preserving the overall trend. This is suitable for medium-term pattern recognition in financial data. The selected time period spans from January 2023 to January 2025, which includes over 100 time points per stock and allows the model to capture recent market conditions across different economic cycles and policy environments.

To ensure that each industry is equally represented, a maximum of 40 stocks were selected from each sector, resulting in a dataset of no more than 160 stocks. Stocks with incomplete historical data or too few valid observations (less than 2) were excluded to maintain data quality. After cleaning, each stock retained around 100 weekly data points, which is sufficient for time-series analysis.

In the preprocessing stage, all closing price sequences were standardized and aligned to a consistent format. This ensured comparability between stocks and eliminated scale differences, helping to improve the accuracy and robustness of distance-based clustering methods.

4.2 Clustering using Soft-DTW

This study applies Soft-DTW as the distance metric to cluster the weekly closing price time series. Compared to traditional DTW, Soft-DTW is differentiable and therefore more suitable for use in optimization frameworks. The number of clusters was set to $k = 4$ to preserve representation across different industries while achieving meaningful segmentation of time series patterns.

To investigate the impact of the smoothing parameter γ on clustering performance, several values (0.001, 0.01, 0.02, and 0.1) were tested. To evaluate clustering performance, three metrics were utilized: the Silhouette Score, the Within-Cluster Sum of Squares (WSS), and the Davies–Bouldin Index (DBI). The Silhouette Score measures how similar a sample is to its own cluster compared to other clusters, with values ranging from -1 to 1, where higher scores indicate better-defined clusters. WSS represents the total squared distance between each point and its assigned cluster center, with lower values implying tighter and more compact clusters. DBI quantifies the average similarity between each cluster and its most similar cluster, where lower values indicate better separation between clusters.

As shown in Figure 4.1, each cluster presents a distinct temporal pattern. Cluster 1 captures sequences with a relatively stable trend until the final segment, where a sharp upward movement is observed. Cluster 2 is characterized by localized fluctuations and subsequent stabilization. Cluster 3 includes series with higher volatility and intermittent peaks, while Cluster 4 exhibits a steady downward trajectory with low variance. These differences highlight the ability of Soft-DTW to effectively identify diverse dynamic behaviors in financial time-series data.

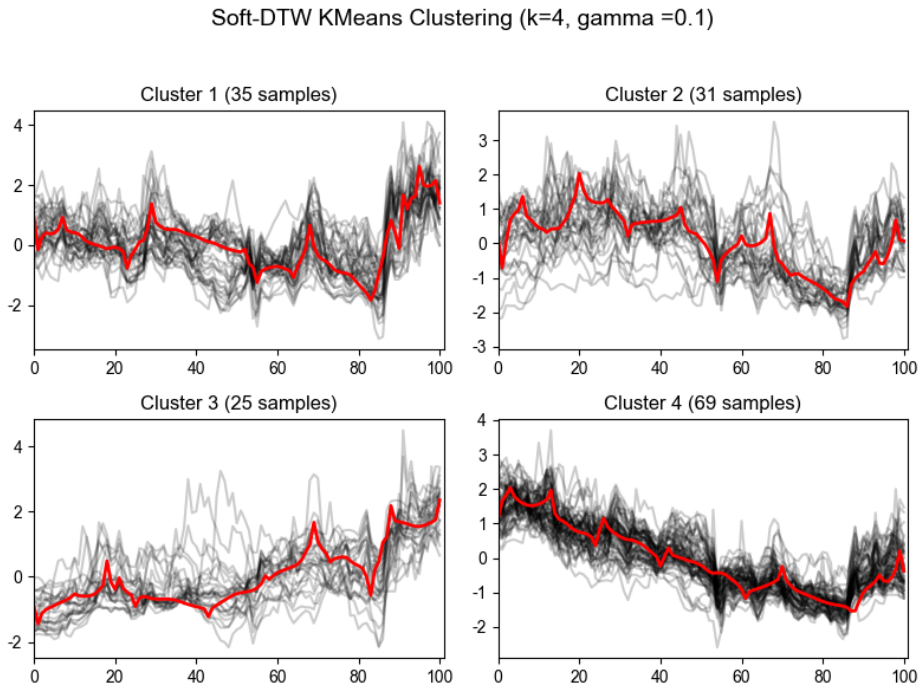

Figure 4.1: Soft-DTW clustering results with $\gamma = 0.1$ visualized using standardized closing prices. Each subplot displays the stock time series within a cluster (black) and the cluster centroid (red).

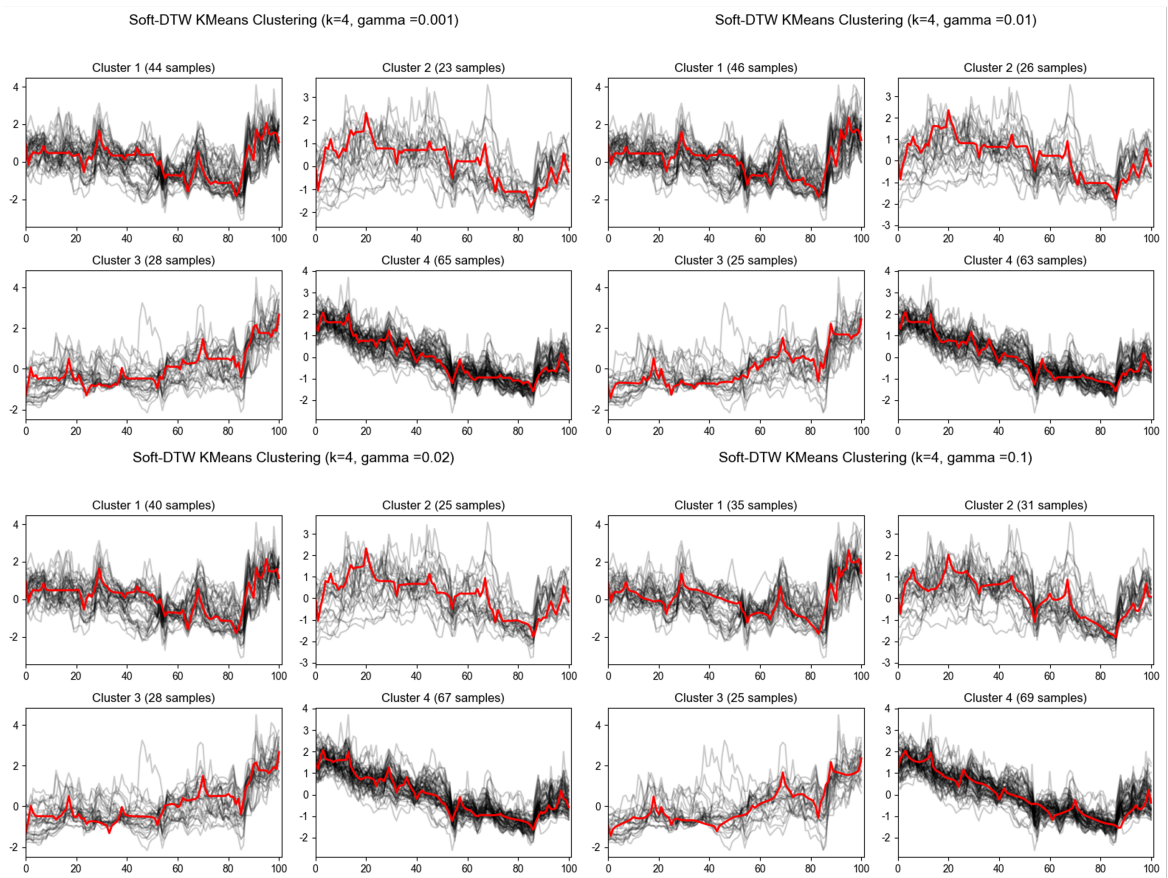

Figure 4.2: Comparison of Soft-DTW clustering results with different γ values. As γ increases, cluster centroids become smoother.

As illustrated in Figure 4.2, lower values of γ (e.g., 0.001) produce cluster centroids that closely follow local noise and fluctuations, potentially leading to overfitting. In contrast, higher values of γ (e.g., 0.1) yield smoother centroids, which better capture the overall trend but may lose sensitivity to small-scale variations.

Table 4.2 quantitatively compares the clustering quality across different γ values. When $\gamma = 0.1$, the clustering achieves the highest Silhouette Score (0.5116), the lowest WSS (305.8795), and the lowest DBI (1.9608), indicating superior cluster compactness and separation. Therefore, $\gamma = 0.1$ was selected for the final analysis.

γ	Silhouette Score	WSS	DBI
0.001	0.4554	1505.6282	2.0324
0.01	0.4521	1497.2387	1.9976
0.02	0.4679	1412.1399	2.0177
0.1	**0.5116**	**305.8795**	**1.9608**

Table 4.2: Comparison of clustering metrics under different γ values.

Overall, the combination of visual inspection and quantitative metrics confirms that a moderate smoothing parameter ($\gamma = 0.1$) provides the best balance between capturing meaningful trends and avoiding overfitting to noise, thereby enhancing the robustness of Soft-DTW-based time-series clustering.

4.3 Cluster-wise Industry Distribution

This section aims to analyze the relationship between clustering results and industry composition, examining whether certain industries dominate specific clusters or whether the clusters exhibit mixed-sector characteristics.

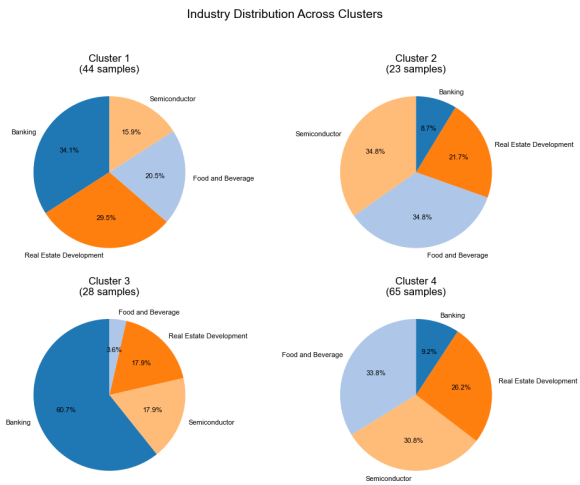

Figure 4.3: Industry composition within each cluster based on Soft-DTW clustering with $\gamma = 0.1$.

As shown in Figure 4.3, the clustering results under $\gamma = 0.1$ reveal clear sectoral differences and concentrations, reflecting the diverse temporal behaviors across industries.

Cluster 1 is primarily composed of Banking stocks (34.1%) and Real Estate Development stocks (29.5%), suggesting that these sectors share similar temporal trends, possibly linked to macroeconomic cycles or policy-driven market movements.

Cluster 2 is characterized by an even split between Semiconductor (34.8%) and Food & Beverage (34.8%) sectors, indicating that despite their different economic roles, certain technology and consumer staples stocks exhibit comparable time-series dynamics, such as moderate volatility and market sensitivity.

Cluster 3 shows the highest industry concentration, with Banking stocks overwhelmingly dominating at 60.7%. This strong dominance suggests that banking stocks tend to move synchronously, likely due to their relatively stable business models and uniform exposure to regulatory environments.

Cluster 4 displays a more balanced distribution among Food & Beverage (33.8%), Semiconductor (30.8%), and Real Estate Development (26.2%) sectors. This mixed composition indicates that stocks from different industries can still exhibit common dynamic patterns, such as gradual declining trends or cyclical corrections.

Overall, these results demonstrate that Soft-DTW clustering with $\gamma = 0.1$ effectively captures both intra-sector consistency and cross-sector similarities in time-series patterns. Certain clusters, such as Cluster 3, highlight strong industry-specific cohesion, while others, like Cluster 4, reveal cross-sector alignment of temporal dynamics.

The use of Soft-DTW proves advantageous in this context by being robust to temporal distortions, capturing non-linear shifts, and providing interpretable clustering outputs aligned with industry structures. Furthermore, the ability to adjust the smoothing parameter γ offers a flexible mechanism to explore different levels of similarity granularity, thereby enhancing the understanding of evolving sectoral behaviors in financial markets.

4.4 Discussion of Results

The clustering results obtained using Soft-DTW with $\gamma = 0.1$ reveal several insightful patterns regarding the temporal dynamics of different industries.

First, the Banking sector exhibits a relatively strong internal consistency. Approximately 60.7% of banking stocks are grouped within Cluster 3, indicating highly synchronized stock price movements among most banking institutions. This behavior likely reflects the sector's stable business models, regulatory oversight, and common exposure to macroeconomic factors such as monetary policy adjustments.

Second, the Semiconductor sector shows considerable heterogeneity. Stocks from this sector are

dispersed across Cluster 1 (15.9%), Cluster 2 (34.8%), and Cluster 4 (30.8%). This dispersion suggests that semiconductor companies experienced diverse market dynamics, possibly due to differences in technological advancements, supply chain dependencies, and varying degrees of global market integration.

Third, the Real Estate Development sector demonstrates a moderate level of clustering, with 29.5% of real estate stocks in Cluster 1 and 26.2% in Cluster 4. This distribution indicates that while certain real estate firms exhibited similar declining or cyclical trends, others displayed divergent behaviors, possibly due to varying financial health, geographic market exposures, and sensitivity to policy tightening in the housing sector.

Fourth, the Food and Beverage sector maintains a relatively high internal consistency, although to a lesser extent compared to banking. Approximately 34.8% of Food and Beverage stocks are grouped in Cluster 2, and 33.8% in Cluster 4. This pattern suggests that consumer-driven sectors experienced similar temporal trends, likely influenced by macroeconomic stability, changing consumer preferences, and resilience to external shocks.

Interestingly, cross-industry similarities are also observed. For instance, stocks from the Semiconductor and Food and Beverage sectors co-exist within Cluster 2, indicating that companies from technologically advanced and consumer-focused industries can sometimes share common market dynamics, potentially driven by broader economic or structural trends rather than sector-specific factors.

Overall, the clustering results demonstrate that Soft-DTW effectively captures both intra-industry consistency and inter-industry temporal synchronization. It reveals nuanced relationships in financial time series that might be overlooked by traditional distance measures, thereby providing deeper insights into the evolving dynamics of market sectors under different economic conditions.

Chapter 5

Conclusion

This dissertation explored the application of Dynamic Time Warping (DTW) and its differentiable extension, Soft-DTW, in time-series clustering, focusing on financial market data. The study addressed the need for flexible distance measures that can accommodate temporal distortions common in real-world time series.

A literature review placed this research within existing clustering methodologies. Traditional distance-based methods, elastic measures like DTW, and modern deep learning models were discussed. DTW's strength in handling time-axis misalignments is well established, but its non-differentiability limits its integration into optimization workflows. Soft-DTW, proposed by Cuturi and Blondel (2017), overcame this by introducing a smooth approximation of the minimum operator, making DTW differentiable and better suited for modern machine learning applications.

Both DTW and Soft-DTW were detailed mathematically. Classical DTW computes the optimal alignment via dynamic programming but lacks smoothness. Soft-DTW introduces a smoothing parameter γ, creating a differentiable loss function while preserving DTW' s core advantages. This modification enables Soft-DTW to integrate seamlessly into gradient-based optimization frameworks.

An empirical study was conducted using Soft-DTW-based K-means clustering on Chinese A-share stock data from four industries: Semiconductors, Banking, Real Estate Development, and Food & Beverage. Weekly closing prices from 2023 to 2025 were standardized and clustered into four groups, using $\gamma = 0.1$ to balance sensitivity and numerical stability.

The results revealed several key patterns. The Food and Beverage sector displayed strong internal consistency, with over 55% of its stocks grouped together, suggesting synchronized reactions to market forces. In contrast, the Semiconductor sector exhibited significant internal heterogeneity, reflecting technological differences and varying market exposures. The Banking sector showed a bifurcated structure, with stocks divided mainly between two clusters, possibly differentiating large national banks from smaller regional institutions. The Real Estate Development

sector showed dispersed behavior across clusters, reflecting the sector' s sensitivity to policy and market segmentation.

Interestingly, companies from different industries, such as Seamild Foods (Food and Beverage) and Suzhou Rural Commercial Bank (Banking), were found within the same cluster. This indicates that Soft-DTW captures broader market-driven temporal patterns beyond strict industry boundaries.

The contributions of this study are threefold. First, it demonstrates the practical value of Soft-DTW in clustering financial time series, effectively identifying both intra-sector consistency and cross-sectoral temporal synchronization. Second, it provides an empirical validation of Soft-DTW' s applicability on real financial data beyond synthetic benchmarks. Third, it offers insights into the sectoral dynamics of the Chinese stock market, enriching the understanding of financial time-series behavior.

Nonetheless, several limitations exist. The dataset was restricted to four sectors and a limited time frame, potentially affecting the generalizability of findings. The choice of the smoothing parameter γ was based on empirical selection rather than systematic optimization. Moreover, Soft-DTW retains the quadratic time complexity of classical DTW, limiting its scalability to larger datasets. Finally, clustering methods are inherently exploratory and do not establish causal relationships, suggesting the need for further confirmatory analyses.

Future research could expand the scope by incorporating more diverse datasets, including international markets and multi-modal indicators. Adaptive methods to optimize the smoothing parameter dynamically could enhance performance. Computational improvements, such as approximate Soft-DTW algorithms, could address scalability concerns. Furthermore, combining Soft-DTW with supervised learning models may open new possibilities for classification, anomaly detection, and predictive modeling in time-series domains.

In conclusion, this dissertation confirms the value of Soft-DTW as a flexible and powerful tool for time-series clustering. By reconciling the robustness of DTW with differentiability, Soft-DTW bridges traditional distance-based approaches with modern machine learning techniques. Its successful application to financial data underscores its potential to uncover meaningful temporal structures, offering significant implications for dynamic data analysis in finance and beyond.

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